PRODUCT FORMULA FOR THE ELEMENTARY TOP COEFFICIENT OF A NATURAL UNIT INTERVAL GRAPH (ORIGINAL THEOREM3.11)

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Editorial note. One count only: exactv1 Theorem3.11. All movable-entry lemmas, involution and fixed-point induction are uncounted supporting current author core.

Editorial note. Selected math is exact arXivv1, not the later revised ALCO version. Historical conjecture wording in the v1 introduction is retained as historical source text.

Editorial note. No author proof was supplied by the assistant. The complete selected argument uses the original prior Schur expansion and dual Jacobi–Trudi identity. The omitted routine-compositions discussion in unrelated Theorem3.16 is outside selected assembly and dependencies.

Editorial note. Author source slips such as Lemma3.7(a), Fix phi_G rather than phi_G-prime in a sentence, and the original induction exponent are retained without mathematical correction.

Editorial note. Inline author TikZ/picture/tableau graphics are retained. Only presentation counters and a standard-package wrapper were added; the biblatex backend is changed from bibtex to biber. Original source/bibliography remain saved.

Editorial note. r002 technical bibliography repair only: in markedgraphs, replace comma-separated author-list syntax by BibTeX and separators for Aliste-Prieto, J.; de Mier, A.; Orellana, R.; Zamora, J. All four authors, entry key, title, arXiv2202.11787 and year2022 are retained. Original references.bib and preamble.tex remain unchanged; task-r001.json preserves the prior task. No mathematical source or bibliography entry is removed.

Editorial note. Presentation-only layout revision: preserve all author mathematics and full front title; allow paragraph stretch or use a short running title to prevent verified clipping.

Editorial scope. One main coefficient product formula, with complete current involution and weighted fixed-point induction. Faithful collection only, not author mathematical certification.

Allowed standard prerequisites (editorial).

- Shareshian–Wachs Schur expansion for natural unit interval graphs. Original Theorem3.2, cited chromposquasi Theorem6.3, complete statement retained at native687–695. Prior q-weighted G-tableau expansion is accepted as the established input; current hook involution and fixed-point classification are fully retained.
- Dual Jacobi–Trudi identity and ordinary symmetric-function basis conventions. Native706–723, cited Sagan Theorem4.5.1 and Stanley Corollary7.16.2; original determinant and hook reduction retained. Standard

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basis identity. No current movable-entry or induction obligation is delegated to it.

- Earlier acyclic-orientation coefficient identity. Original native401–422 cites Stanley Theorem3.3 and Shareshian–Wachs Theorem5.3. Context for §3 first sentence only; the selected proof proceeds through author tableaux, not the later unrelated orientation bijection.

1. Introduction

Let $G = (V, E)$ be a graph with vertex set $V = \{v_1, \ldots, v_n\}$ and edge set E. Also let $\boldsymbol{x} = \{x_1, x_2, \ldots\}$ be a set of commuting variables indexed by the positive integers $\mathbb{N}$. In his groundbreaking 1995 work, Stanley defined the *chromatic symmetric function* [19, Definition 2.1]

$$X_G(\boldsymbol{x}) = \sum_{\kappa} x_{\kappa(v_1)} \cdots x_{\kappa(v_n)},$$

where the sum is over all *proper colourings* of G, which are functions $\kappa : V \to \mathbb{N}$ such that whenever $ij \in E$, we have $\kappa(i) \neq \kappa(j)$. Consider the expansion

$$X_G(\boldsymbol{x}) = \sum_{\lambda} c_\lambda e_\lambda \tag{1.1}$$

in the basis of *elementary symmetric functions*. We reserve the notation c_λ for these coefficients. We say that $X_G(\boldsymbol{x})$ is *e-positive* if all the c_λ are nonnegative, that is, every e_λ which appears does so with positive coefficient. Stanley and Stembridge [19, Conjecture 5.1], [21, Conjecture 5.5] made the following conjecture which has become one of the driving forces behind the study of $X_G(\boldsymbol{x})$.

Conjecture 1.1 ($(\mathbf{3}+\mathbf{1})$-free Conjecture)**.** If G is the incomparability graph of a $(\mathbf{3}+\mathbf{1})$-free poset, then $X_G(\boldsymbol{x})$ is e-positive.

Guay-Paquet [12, Theorem 5.1] showed that it suffices to prove e-positivity whenever G is the incomparability graph of a poset which is both $(\mathbf{3}+\mathbf{1})$- and $(\mathbf{2}+\mathbf{2})$-free. These are called *unit interval graphs*. Shareshian and Wachs defined a q-analogue of $X_G(\boldsymbol{x})$ [18, Definition 1.2] as follows. Suppose G has vertex set $V = [n] := \{1, \ldots, n\}$. Now a proper coloring $\kappa : [n] \to \mathbb{N}$ has *ascent number*

$$\operatorname{asc} \kappa = \#\{ij \in E \mid i < j \text{ and } \kappa(i) < \kappa(j)\},$$

where we will use a hash tag or a pair of vertical bars to denote cardinality. Now define the *chromatic quasisymmetric function* to be

$$X_G(\boldsymbol{x}; q) = \sum_{\kappa} q^{\operatorname{asc}(\kappa)} x_{\kappa(1)} \cdots x_{\kappa(n)},$$

summed over proper $\kappa : [n] \to \mathbb{N}$. To generalize Conjecture 1.1 to this setting, it suffices to consider unit interval graphs with a particular labeling. Say that $G = ([n], E)$ is a *natural unit interval graph* if for all $1 \leq i < j < k \leq n$ we have

$$ik \in E \text{ implies } ij \in E \text{ and } jk \in E. \tag{1.2}$$

Although $X_G(\boldsymbol{x}; q)$ is not generally symmetric in the $\boldsymbol{x}$ variables, Shareshian and Wachs [18, Theorem 4.5] showed that $X_G(\boldsymbol{x}; q)$ is symmetric whenever G is a natural unit interval graph. So we have an elementary symmetric function expansion

$$X_G(\boldsymbol{x}; q) = \sum_{\lambda} c_\lambda(q) e_\lambda \tag{1.3}$$

where the $c_\lambda(q)$ are polynomials in q. Again, the notation $c_\lambda(q)$ will always refer to these coefficients. Call the expansion *e-positive* if the coefficients in each $c_\lambda(q)$ are nonnegative. This leads to the following conjecture.



$X_G(\boldsymbol{x}) = 4e_{32} + 12e_{41} + 20e_5$ $\qquad$ $X_H(\boldsymbol{x}) = e_{211} - 2e_{22} + 5e_{31} + 4e_4$

FIGURE 1. The bowtie graph, the claw graph, and their chromatic symmetric functions

Conjecture 1.2 (Shareshian-Wachs)**.** If G is a natural unit interval graph then $X_G(\boldsymbol{x}; q)$ is e-positive.

Note that when we set $q = 1$ we recover Conjecture 1.1, so $c_\lambda(1) = c_\lambda$. Several authors proved e-positivity for particular classes of natural unit interval graphs G. Harada and Precup [13, Theorem 6.1] proved that $X_G(\boldsymbol{x}; q)$ is e-positive whenever G has independence number 2 using cohomology of abelian Hessenberg varieties and Cho and Huh [4, Theorem 3.3] gave a purely combinatorial proof. Cho and Hong [3, Theorem 1.8] proved that $X_G(\boldsymbol{x})$ is e-positive whenever G has independence number 3. Dahlberg [7, Corollary 5.4] proved that $X_G(\boldsymbol{x})$ is e-positive whenever G has clique number 3. Gebhard and Sagan [9, Corollary 7.7] proved that $X_G(\boldsymbol{x})$ is e-positive whenever G is formed by joining a sequence of cliques at single vertices and Tom [22, Corollary 4.20] proved that $X_G(\boldsymbol{x}; q)$ is e-positive for such graphs by finding an explicit formula.

One can alternatively take a dual approach to the Stanley–Stembridge conjecture by fixing a particular partition λ and proving nonnegativity of $c_\lambda(q)$ or c_λ for all natural unit interval graphs G. Hwang [14, Theorem 5.13] proved that $c_\lambda(q)$ is nonnegative whenever λ is a hook. Abreu and Nigro [1, Corollary 1.10] and independently Rok and Szenes [15, Theorem 4.1] proved that c_λ is nonnegative whenever λ has exactly two rows. Clearman, Hyatt, Shelton and Skandera [6, Theorem 10.3] proved that $c_\lambda(q)$ is nonnegative whenever λ has exactly two columns, using a connection to characters of Hecke algebras.

Recall that an *acyclic orientation* of $G = ([n], E)$ is an assignment O of directions to each edge of G such that there are no directed cycles. Directed edges are also called *arcs* and an arc from i to j is denoted $i \to j$. Vertex i is a *sink* of O if there are no outgoing arcs from i. The *ascent set* of O is

$$\operatorname{Asc} O = \{i \to j \mid i < j\}$$

with ascent number

$$\operatorname{asc} O = \#\operatorname{Asc} O.$$

Stanley [19, Theorem 3.3] proved that for an arbitrary graph G, the sum

$$a_j := \sum_{\lambda:\ \lambda'_1 = j} c_\lambda \tag{2.4}$$

is the number of acyclic orientations of G with exactly j sinks. Shareshian and Wachs [18, Theorem 5.3] proved that if G is a natural unit interval graph, we have

$$a_j(q) := \sum_{\lambda:\ \lambda'_1 = j} c_\lambda(q) = \sum_{O:\ \text{exactly } j \text{ sinks}} q^{\operatorname{asc} O}. \tag{2.5}$$



FIGURE 4. The four G-tableaux of shape $\lambda = 221$ for the bowtie graph G

Let us say that vertices u and v of a graph $G = ([n], E)$ are *equivalent* if they have the same closed neighbourhood. Note that equivalent vertices must be adjacent. We now prove a divisibility result about the coefficients c_λ and $c_\lambda(q)$. For an integer k we define the standard q-analogues

$$[k]_q = 1 + q + q^2 + \cdots + q^{k-1} = \frac{q^k - 1}{q - 1}$$

and

$$[k]_q! = [k]_q [k-1]_q \cdots [2]_q [1]_q.$$

3. The Schur basis

If G is a natural unit interval graph then equation (2.5) gives an interpretation of the coefficient of e_n in $X_G(\boldsymbol{x}; q)$ as a generating function for the acyclic orientations of G with one sink. In this section, we use the Schur basis to give another interpretation for the coefficient of e_n in terms of tableaux. Throughout this section $G = ([n], E)$ will be a natural unit interval graph. Gasharov [8, Theorem 4] proved a combinatorial formula for the Schur expansion of $X_G(\boldsymbol{x})$ in terms of certain tableaux and Shareshian and Wachs [18, Theorem 6.3] proved a q-analogue of this result. We will write our Young diagrams in English notation.

Definition 3.1. A *G-tableau of shape λ* is a bijective filling T of the diagram of λ with the numbers 1 through n such that the following two conditions hold.

(C1) If an entry i is directly to the left of an entry j in the same row, then $i < j$ and $ij \notin E$.
(C2) If an entry i is directly above an entry j in the same column, then either $i < j$ or $ij \in E$.

Let GTab_λ denote the set of G-tableaux of shape λ. An *inversion* of T is a pair of entries (i, j) such that

(I1) $i < j$,
(I2) $ij \in E$, and
(I3) i is in a lower row than j.

We denote by $\operatorname{Inv} T$ the set of inversions of T and $\operatorname{inv} T = \#\operatorname{Inv} T$.

Theorem 3.2 ([18])**.** The chromatic quasisymmetric function of a natural unit interval graph G satisfies

$$X_G(\boldsymbol{x}; q) = \sum_\lambda \sum_{T \in \mathrm{GTab}_\lambda} q^{\operatorname{inv} T} s_\lambda. \tag{3.1}$$

In particular, $X_G(\boldsymbol{x}; q)$ is Schur-positive.

Example 3.3. Let G be the bowtie graph from Figure 1. The four G-tableaux of shape $\lambda = 221$ and their inversion sets are given in Figure 4. By Theorem 3.2, the coefficient of s_{221} in $X_G(\boldsymbol{x}; q)$ is $(q^2 + 2q^3 + q^4)$.

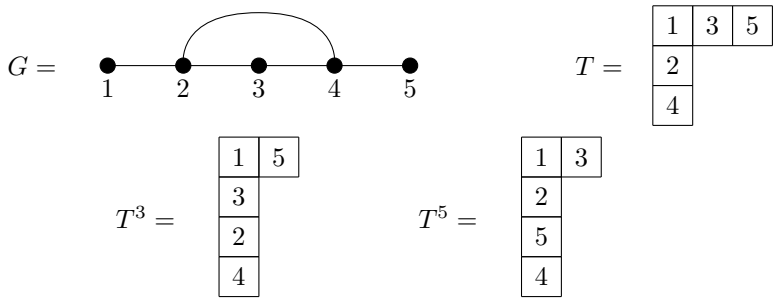


FIGURE 5. A natural unit interval graph G, a G-tableau T, and two T^j

Now we can calculate coefficients in the elementary basis by first using Theorem 3.2 and then converting Schur functions to elementary symmetric functions using the dual Jacobi–Trudi determinantal identity [16, Theorem 4.5.1], [20, Corollary 7.16.2],

$$s_\lambda = \det(e_{\lambda'_i - i + j})_{i,j=1}^{\lambda_1}. \tag{3.2}$$

In particular, the coefficient of e_n in s_λ is nonzero only if λ is a *hook*, meaning a partition of the form $\lambda = k1^{n-k}$ for some $1 \leq k \leq n$, in which case this coefficient is $(-1)^{k-1}$. Let

$$\mathrm{HT}(G) = \{T \mid T \text{ is a } G\text{-tableau of hook shape}\},$$

and if $T \in \mathrm{HT}(G)$ has shape $k1^{n-k}$ then define its *sign* to be

$$\operatorname{sgn} T = (-1)^{k-1}.$$

By Theorem 3.2 and (3.2), the coefficient of e_n in $X_G(\boldsymbol{x}; q)$ is

$$c_n(q) = \sum_{T \in \mathrm{HT}(G)} (\operatorname{sgn} T) q^{\operatorname{inv} T}. \tag{3.3}$$

We now define a sign-reversing involution φ_G on $\mathrm{HT}(G)$ to obtain a positive combinatorial description for $c_n(q)$. The idea will be to move entries between the *arm* and *leg* of T, which are the first row and first column of T, respectively, excluding the corner entry in the $(1,1)$-cell. We will only move entries if we can preserve the inversion set.

Definition 3.4. Let $T \in \mathrm{HT}(G)$ be a G-tableau. An entry j in the arm of T is *movable* if it can be moved to the leg of T to produce a tableau T^j such that

(M1) $T^j \in \mathrm{HT}(G)$, and
(M2) $\operatorname{Inv} T^j = \operatorname{Inv} T$.

Similarly, entry j is *movable* from the leg to the arm if the resulting tableau T^j satisfies (M1) and (M2).

We will see shortly that if j is movable then the position to which j can be moved, and hence T^j, are uniquely determined.

Example 3.5. In the first line of Figure 5 we see a natural unit interval graph, G, and one of its G-tableaux, T, of hook shape. It is easy to check that $\operatorname{Inv} T = \{(2,3),(4,5)\}$. Both 3 and 5 are movable and the resulting G-tableaux are displayed in the second line. The elements 2 and 4 are not movable.

Lemma 3.6. Let $T \in \mathrm{HT}(G)$ be a G-tableau.

(a) An entry j in the leg of T is movable if and only if every entry i of T in a row above j satisfies $ij \notin E$.
(b) If the entry j is movable, there is a unique position to which it can be moved and every entry i above j in the leg of T satisfies $i < j$.

Proof. (a) If j is movable, then we must have $ij \notin E$ for every entry i in the first row of T by (C1) and because G is a natural unit interval graph. We also must have $ij \notin E$ for every entry i in the leg of T above j because otherwise there would be a change in inversions.

Conversely, suppose that $ij \notin E$ for every entry i above j and let T^j be the tableau obtained by moving j to the arm so that it is strictly increasing, so (C1) holds. We have $\operatorname{Inv} T^j = \operatorname{Inv} T$ because the only change in relative positions occurred between non-adjacent vertices of G. If there are entries x and y respectively directly above and below j in T, we cannot have $x > y$ and $xy \notin E$ because then either $j > x > y$ and $yj \notin E$ because G is a natural unit interval graph, violating (C2), or $j < x$ and $xj \notin E$ by assumption, again violating (C2). Therefore, the tableau T^j is indeed a G-tableau and the entry j is movable.

(b) First note that uniqueness holds because the first row of T^j must be strictly increasing by (C1). Next, suppose that j is movable and there is an entry $i > j$ above j in the leg of T. We have shown that $ij \notin E$ so i must not be directly above j by (C2). Therefore, there is some other entry i' directly above j, and $i'j \notin E$ because j is movable, so $i' < j$. But now because $i > j > i'$, there must be entries x and y in the leg of T above j with x directly above y and $x > j > y$. But because j is movable we have $jx \notin E$, and $xy \notin E$ because G is a natural unit interval graph, which violates (C2). □

Lemma 3.7. Let $T \in \mathrm{HT}(G)$ be a G-tableau. Every entry in the arm of T is movable and can be moved to a unique position.

Proof. Let j be an entry in the arm of T and let $x_1, \ldots, x_\ell$ be the entries of the first column of T from top to bottom. By (C1) and the fact that G is a natural unit interval graph, we have $x_1 < j$ and $x_1 j \notin E$. Therefore, there is some maximal index i with $1 \leq i \leq \ell$ such that we have $x_t < j$ and $x_t j \notin E$ for every $1 \leq t \leq i$. We claim that the entry j is movable to the position directly below the entry x_i. The first row of the resulting tableau T^j satisfies (C1) because G is a natural unit interval graph. We have $x_i < j$ by definition and we cannot have both $x_{i+1} < j$ and $x_{i+1} j \notin E$ by maximality of i, so the first column of T^j satisfies (C2). We have $\operatorname{Inv} T^j = \operatorname{Inv} T$ because the entries in the arm of T^j and the entries $x_1, \ldots, x_i$ that are now above j are not adjacent to j so there is no change in inversions.

Finally, the entry j could not have moved to a higher position because (C2) would not hold by definition of i. Also, the entry j could not have moved to a lower position because then it would be movable in T^j, which means by Lemma 3.6 that $x_{i+1} < j$ and $x_{i+1} j \notin E$, contradicting maximality of i. □

Note that by Lemma 3.6 and Lemma 3.7, movable entries can be moved to a unique position so the tableau T^j is well-defined, as promised.

Definition 3.8. Let $T \in \mathrm{HT}(G)$ be a G-tableau. We define the G-tableau

$$\varphi_G(T) = \begin{cases} T & \text{if } T \text{ has no movable entry,} \\ T^j & \text{if } T \text{ has smallest movable entry } j. \end{cases} \tag{3.4}$$

We denote by $\operatorname{Fix} \varphi_G$ the set of *fixed points* of φ_G, meaning those tableaux T for which $\varphi_G(T) = T$.

We now show that the map $\varphi_G : \mathrm{HT}(G) \to \mathrm{HT}(G)$ is an inv-preserving, sign-reversing involution.

Lemma 3.9. We have $\varphi_G(\varphi_G(T)) = T$ and $\operatorname{inv}\varphi_G(T) = \operatorname{inv} T$ for every $T \in \mathrm{HT}(G)$. Also

$$\operatorname{sgn}\varphi_G(T) = \begin{cases} 1 & \text{if } T \in \operatorname{Fix}\varphi_G, \\ -\operatorname{sgn} T & \text{else.} \end{cases}$$

Proof. This first statement is clear if T has no movable entry because $\varphi_G(T) = T$, so suppose that T has smallest movable entry j, so that $\varphi_G(T) = T^j$. We have $\operatorname{inv} T^j = \operatorname{inv} T$ by definition of φ_G. By construction, the entry j of T^j is movable and the unique position to which it can be moved was its original position in T. However, we need to check that the entry j of T^j is in fact the smallest movable entry. We will show that if a smaller entry $i < j$ of T^j is movable, then i was movable in T, contradicting minimality of j.

By Lemma 3.7 (a), every entry in the arm of a hook tableau is movable. So if i were in the arm of T^j then it would have been movable in T. Now suppose that i is in the leg of T^j. By Lemma 3.6, the condition for i to be movable is that it is not adjacent to any entries in a higher row. So i movable in T^j but not T means that it is not adjacent to everything above it in T^j but adjacent to something above in T. This can only happen if j is in the arm of T, moves to the leg of T^j below i, and $ij \in E$. But j being movable in T^j implies $ij \notin E$ which is a contradiction.

Finally, if $T \in \operatorname{Fix}\varphi_G$, then T has no movable entry. By Lemma 3.7, there are no elements in the arm so that T has shape 1^n and $\operatorname{sgn} T = 1$. If $T \notin \operatorname{Fix}\varphi_G$, then φ_G changes the number of columns in passing from T to $\varphi_G(T)$ by exactly one. It follows that $\operatorname{sgn}\varphi_G(T) = -\operatorname{sgn} T$ which completes the proof. □

Corollary 3.10. Let G be a natural unit interval graph. Then the coefficient of e_n in $X_G(\boldsymbol{x}; q)$ is given by

$$c_n(q) = \sum_{T \in \operatorname{Fix}\varphi_G} q^{\operatorname{inv} T}. \tag{3.5}$$

Proof. By Lemma 3.9, the map φ_G is an inversion-preserving bijection between the positively-signed non-fixed points and the negatively-signed non-fixed points, so the sum in (3.3) is equal to the sum over the fixed points, which are positively signed. □

We now describe the fixed points of φ_G to get an explicit formula for $c_n(q)$ which will follow from Corollary 3.14 below.

Theorem 3.11. Let $G = ([n], E)$ be a natural unit interval graph. For $2 \le k \le n$, let b_k denote the number of smaller neighbours of vertex k in G. Then the coefficient of e_n in the chromatic quasisymmetric function $X_G(\boldsymbol{x}; q)$ is

$$c_n(q) = [n]_q[b_2]_q[b_3]_q \cdots [b_n]_q. \tag{3.6}$$

By considering the reverse graph obtained by relabeling vertex k to $n - k + 1$, we could equivalently think about the number of larger neighbours of each vertex.

Example 3.12. For the bowtie graph G from Figure 1, the coefficient of e_5 in $X_G(\boldsymbol{x}; q)$ is

$$c_5(q) = [5]_q[2]_q[2]_q = (1 + q + q^2 + q^3 + q^4)(1 + q)^2. \tag{3.7}$$

We will use induction on n to enumerate the fixed points of φ_G. Suppose that $n \ge 2$ and let G' denote the graph G with the vertex n removed. For a tableau T of column shape, we denote by T_i the i-th entry from the top.

Lemma 3.13. If $b_n = 0$, then $\operatorname{Fix}\varphi_G = \emptyset$. Otherwise, the tableaux $T \in \operatorname{Fix}\varphi_G$ arise precisely by taking a tableau $T' \in \operatorname{Fix}\varphi_{G'}$ and inserting the entry n either at

the bottom, directly above a neighbour of n other than the one occurring highest in T', or at the top if T'_1 is a neighbour of n.

Proof. If $b_n = 0$, then a G-tableau $T \in \mathrm{GTab}_{1^n}$ must have the entry n at the bottom to satisfy (C2). But then the n is movable by Lemma 3.6. So assume that $b_n \geq 1$. A tableau T constructed from T' in one of the three given ways will satisfy (C2) because n is either at the bottom or directly above a neighbour, and will have no movable elements by Lemma 3.6 because n is either at the top or below a neighbour, so we have $T \in \operatorname{Fix} \varphi_G$. We now show that a tableau $T \in \operatorname{Fix} \varphi_G$ must arise in this way.

Let T' denote the tableau obtained from T by removing the entry n. We must first show that T' still satisfies (C2). Suppose, to the contrary, that there are entries x and y respectively above and below the entry n in T with $x > y$ and $xy \notin E$. But since G is a natural unit interval graph this would force $yn \notin E$ since $n > y$. So T would not satisfy (C2), a contradiction.

Also we claim that T' has no movable elements. Suppose that T' has a movable element j. By Lemma 3.6, every element i above j in T' must satisfy $i < j$ and $ij \notin E$. Because j is not movable in T, the entry n must be above j in T and $jn \in E$ by Lemma 3.6 (a). Now every entry i above n in T satisfies $i < j < n$ which implies $in \notin E$ because G is a natural unit interval graph. So, by Lemma 3.6 (a) again, the only way for n to not be movable in T is if $T_1 = n$. If $T_2 = j$, then $T'_1 = j$ so j is not movable in T'. Otherwise, T_2 must be some other entry k. However, because k is above j in T', we have shown that $k < j$ and $kj \notin E$. This forces $k < n$ and $kn \notin E$ so that T does not satisfy (C2). This final contradiction shows that no element of T' is movable.

Therefore $T' \in \operatorname{Fix} \varphi_G$ and indeed T arises by inserting the entry n somewhere in T'. In order to satisfy (C2), the entry n must be inserted either at the bottom or directly above a neighbour of n, and in order for n to not be movable, by Lemma 3.6, it must be either at the top or have a neighbour above it. □

Theorem 3.11 follows from the next corollary by summing over $1 \leq j \leq n$.

Corollary 3.14. Let $G = ([n], E)$ be a natural unit interval graph. Then for every $1 \leq j \leq n$, we have

$$\sum_{T \in \operatorname{Fix} \varphi_G,\ T_1 = j} q^{\operatorname{inv} T} = q^{j-1}[b_2]_q \cdots [b_n]_q. \tag{3.8}$$

Proof. We use induction on n, the case of $n = 1$ trivial, so suppose that $n \geq 2$. If $b_n = 0$, then the statement holds by Lemma 3.13, so assume that $b_n \geq 1$ and note that because G is a natural unit interval graph, the neighbours of n are exactly $n - b_n, \ldots, n - 1$. If $j < n$, then by Lemma 3.13, every tableau $T \in \operatorname{Fix} \varphi_G$ with $T_1 = j$ arises from a tableau $T' \in \operatorname{Fix} \varphi_{G'}$ by inserting the entry n in one of b_n possible positions. Inserting n in the i-th place from the bottom results in $(i-1)$ new inversions, so by our induction hypothesis, we have

$$\sum_{T \in \operatorname{Fix} \varphi_G,\ T_1 = j} q^{\operatorname{inv} T} = (1 + q + \cdots + q^{b_n - 1}) \sum_{T' \in \operatorname{Fix} \varphi_{G'},\ T'_1 = j} q^{\operatorname{inv} T} = q^{j-1}[b_2]_q \cdots [b_n]_q.$$

If $j = n$, then by Lemma 3.13, every tableau $T \in \operatorname{Fix} \varphi_G$ with $T_1 = n$ arises from a tableau $T' \in \operatorname{Fix} \varphi_G$ with $n - b_n \leq T'_1 \leq n - 1$ by inserting the entry n at the top.

Doing so produces b_n new inversions, so by our induction hypothesis, we have

$$\sum_{T\in \operatorname{Fix}\varphi_G,\ T_1=n} q^{\operatorname{inv} T} = \sum_{i=n-b_n}^{n-1} \sum_{T'\in \operatorname{Fix}\varphi_{G'},\ T'_1=i} q^{\operatorname{inv} T+b_n}$$
$$= q^{b_n} \sum_{i=n-b_n}^{n-1} q^{i-1}[b_2]_q\cdots[b_{n-1}]_q = q^{n-1}[b_2]_q\cdots[b_n]_q,$$

which completes the proof. □

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References

[1] A. Abreu and A. Nigro. "Splitting the cohomology of Hessenberg varieties and e-positivity of chromatic symmetric functions". In: (2023). arXiv: 2304.10644.

[2] J. Aliste-Prieto et al. "Marked graphs and the chromatic symmetric function". In: (2022). arXiv: 2202.11787.

[3] S. Cho and J. Hong. "Positivity of chromatic symmetric functions associated with Hessenberg functions of bounce number 3". In: *Electron. J. Combin.* 29.2.19 (2022).

[4] S. Cho and J. Huh. "On e-positivity and e-unimodality of chromatic quasisymmetric functions". In: *SIAM J. Discrete Math.* 33.4 (2019).

[5] S. Cho and S. van Willigenburg. "Chromatic bases for symmetric functions". In: *Electron. J. Combin.* 23 (2016).

[6] S. Clearman et al. "Evaluation of Hecke algebra traces at Kazhdan–Lusztig basis elements". In: *Electron. J. of Combin.* 23 (2016).

[7] S. Dahlberg. "Triangular ladders $P_{d,2}$ are e-positive". In: (2018). arXiv: 1811.04885.

[8] V. Gasharov. "Incomparability graphs of $(3+1)$-free posets are s-positive". In: *Discrete Math.* 157 (1996).

[9] D. Gebhard and B. Sagan. "A chromatic symmetric function in noncommuting variables". In: *J. Algebr. Comb.* 13 (2001).

[10] C. Greene and T. Zaslavsky. "On the interpretation of Whitney numbers through arrangements of hyperplanes, zonotopes, non-Radon partitions, and orientations of graphs". In: *Trans. Am. Math. Soc.* 280.1 (1983).

[11] Curtis Greene. "An extension of Schensted's theorem". In: *Advances in Math.* 14 (1974), pp. 254–265. ISSN: 0001-8708. DOI: 10.1016/0001-8708(74)90031-0. URL: https://doi.org/10.1016/0001-8708(74)90031-0.

[12] M. Guay-Paquet. "A modular law for the chromatic symmetric functions of $(3+1)$-free posets". In: (2013). arXiv: 1306.2400.

[13] M. Harada and M. Precup. "The cohomology of abelian Hessenberg varieties and the Stanley–Stembridge conjecture". In: *Algebr. Comb.* 2 (2019).

[14] B.-H. Hwang. "Chromatic quasisymmetric functions and noncommutative P-symmetric functions". In: (2022). arXiv: 2208.09857.

[15] A. Rok and A. Szenes. "Eschers and Stanley's chromatic e-positivity conjecture in length-2". In: (2023). arXiv: 2305.00963.

[16] B. Sagan. *The Symmetric Group: Representations, Combinatorial Algorithms, and Symmetric Functions*. Springer New York, 2001.

[17] C. Schensted. "Longest increasing and decreasing subsequences". In: *Canadian J. Math.* 13 (1961), pp. 179–191. ISSN: 0008-414X,1496-4279. DOI: 10.4153/CJM-1961-015-3. URL: https://doi.org/10.4153/CJM-1961-015-3.

[18] J. Shareshian and M. Wachs. “Chromatic quasisymmetric functions”. In: *Adv. Math.* (2016).

[19] R. Stanley. “A symmetric function generalization of the chromatic polynomial”. In: *Adv. Math.* 111 (1995).

[20] R. Stanley. *Enumerative Combinatorics. Vol. 2*. Cambridge University Press, 1999.

[21] R. Stanley and J. Stembridge. “On immanants of Jacobi–Trudi matrices and permutations with restricted position”. In: *J. Combin. Theory Ser. A* 62 (1993).

[22] F. Tom. “A signed e-expansion of the chromatic quasisymmetric function”. In: (2024). arXiv: 2110.07984.